

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	CA18-SS-0000000009
Award Program	Pollinator Health
Project Title	
Grant Start Date	Jan 01/2018
Grant End Date	Nov/01/2023
Total Grant Budget	\$ 980,712.00
Total FFAR Budget	\$ 490,356.00
Matching Funder(s)	UCR, UO
Final Report Date	Nov/01/2022
Project Director/Principal Investigator Information	
Full Name	Lauren Ponisio
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Grantee Organization Information	
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Authorized Signing Official (ASO) Information	
Name	Franny Vandervoort

ASO Title	Sponsored Projects Administrator
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Phone	541-346-7475

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Yolo County, California, CA Congressional District 3

Riverside, California, CA Congressional District 41

Eugene, Oregon, OR Congressional District 4

- 1.2. How many new jobs were created because of this grant funding?

14: (4 field technicians, 2 postdoctoral scholars, 2 graduate student research assistantship, 4 lab technicians, 2 undergraduate assistantships)

- 1.3. How many jobs were maintained because of this grant funding?

NA

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No.

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No.

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Our work contributed to our understanding of the health of wild and managed bees in mass

flowering agricultural systems. We developed protocols for the community-wide screening of over a thousand bees for 7 common parasites and 3 RNA viruses. We simultaneously characterized their bacterial gut microbiomes and the pollen on their bodies using Illumina sequencing in order to relate a bee's health to his foraging and gut microbes. Finally, we deployed *Osmia California* nests to understand how pesticide exposure affected offspring production. To our knowledge this is the largest community-wide characterization of the bee parasites, pathogens, bacterial microbes and their interaction patterns and pesticide exposure.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Key finding 1: To understand how mass-flowering crops impact host communities and subsequently amplify or dilute parasitism, we surveyed wild and managed bees in a monoculture landscape with varying degrees of floral diversification. We screened 1509 bees from 16 genera in sunflower fields and in non-crop flowering habitat across 200 km² of the California Central Valley. We found that mass-flowering crops increase bee abundance. Wild bee abundance was subsequently associated with higher parasite presence, but only in sites with a low abundance of non-crop flowers. Bee traits related to higher dispersal ability (body size) and diet breadth (pollen lecty) were also positively related to parasite presence. Our results highlight the importance of non-crop flowering habitat for supporting bee communities. We suggest monoculture alone cannot support healthy bees.

Key findings 2: The microbiome and pathobiome are key determinants of how wildlife responds to global change, but it is unknown why microbial associations vary between and within species. We examined if host traits shape microbial associations at the species-level and individual-level. Using multilevel phylogenetic Bayesian models, we analyzed the microbiomes and parasites of 1,686 wild bees and managed honey bees across 34 species that vary widely in phenotypic traits. We found that bees with more distinct traits host more distinct pathobiomes. For the microbiome, realized and fundamental diet breadth were a stronger predictor of microbial distinctness and diversity than trait distinctness. We also found that microbiome and pathobiome community composition were correlated, and that microbiome composition was further shaped by the community composition of the pollen collected by individual bees. Our findings demonstrate that both species-level and individual-level variation in host traits act as filters affecting the assembly of microbes in free-living organisms.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be

explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Disease has been identified as a primary driver of biodiversity loss and disease risk is exacerbated by habitat loss. Although agriculture occupies half of the earth's habitable land the spread of disease in agriculture is rarely considered for wildlife. To slow global species loss, it is imperative to understand wildlife epidemiology in agriculture. In particular, it is important to address how agriculture changes the availability of resources for wildlife because shifts in resource availability have been shown to alter infectious disease dynamics in wildlife.

As agricultural production continues to intensify, many economically important crops (e.g., nut, fruit, and oilseed crops) are increasingly shifting to 'monoculture'—the cultivation of dense, single species fields. Monocultures are often characterized by synchronized mass-flowering events. These events provide a resource pulse of pollen, nectar, and temporary habitat for wildlife at dramatically high levels, but only for a short duration. How mass-bloom events in monoculture affects infection risk is unknown, but monoculture likely shapes host behavior and abundance, which has been linked to parasite transmission. In agricultural systems, organisms dependent on floral resources will track the spatio-temporal distribution of bloom events within a season. Repeated annual mass-bloom events can increase population sizes and community richness. Hosts whose populations are reliant on resource pulses may be uniquely vulnerable to infection if their parasites are associated with those resources. Mass-bloom events may increase parasitism rates in organisms (amplification) if they result in host aggregation, increasing exposure between individuals. Alternatively, parasitism may decline following a resource pulse if the event attracts hosts who vary in their ability to transmit and become infected, known as dilution. Dilution might also occur if the resources gathered support an effective immune response to parasitism. We found that mass-blooms act to aggregate individuals. Increases to host abundance were subsequently associated with amplified parasitism, likely by increasing exposure and transmission between individuals. We find that by shaping bee population dynamics, mass-bloom events in monoculture increase parasite risk in bees.

Crop plants are not the only floral resource that can impact parasite epidemiology in agriculture. On farms, diversification practices have been adopted that enhance biodiversity and provide ecosystem services. For example, strips of native plants along field edges, called 'hedgerows', can support higher richness and abundance of wildlife including beneficial insects, mammals, and birds. Farms also host a variety of unmanaged, weedy species along field margins that can attract biodiversity. If non-crop resources from hedgerows and weeds aggregate hosts, this may lead to parasite amplification. Alternatively, non-crop resources may dilute parasitism if they increase host species richness or immunity. Whether mass-

blooming crops and non-crop habitat work in synergy to amplify parasitism or whether one negates the effects of the other, has implications for designing agricultural landscapes that support biodiversity. Encouragingly, non-crop floral resources appeared to mitigate parasite amplification. As non-crop floral abundance at a site increased, the positive effect of bee abundance on parasitism diminished such that, at sites with the highest floral abundance, the relationship between bee abundance and parasitism was negative.

We also found that transmission dynamics in bee communities were associated with bee traits. Specifically, larger bees and oligolectic species (pollen specialists) had higher parasite prevalence and richness. Larger bees forage over longer distances and have greater surface area with which to pick up parasites. Our results parallel findings in vertebrates that host traits related to movement ecology and feeding behavior were the most important predictors of how supplemental resources affected parasitism. Simplified landscapes have been shown to favor generalists and smaller bees, likely because these bees have less-specialized resource needs. Our results suggest that increased parasitism may also contribute to their lower persistence in agricultural landscapes. By identifying which bee traits are associated with increased infection risk, we can predict which species are most vulnerable to changes in resource landscapes.

It is unknown whether all mass-flowering crops amplify parasitism, but the fact that we found this effect in sunflower, whose consumption has been shown to reduce parasite infection intensity in bees, suggests that amplification may be a widespread phenomenon. Wild bees are critical for enhancing yields and yield stability, but parasitism may impact population persistence, individual foraging efficiency, and thus pollination services and yields. This particular agricultural landscape is very resource scarce for bees, as the California Central Valley has seen approximately 90% habitat loss and fragmentation since the 1840s. While there are challenges to habitat restoration and other diversification techniques, our work highlights the importance for these practices for mitigating the spread of disease and supporting healthy wildlife populations.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Fieldwork: (Field work 2018-2019, completed). We collected pollinator specimens from 12 sites across Yolo Co. CA. We will sampled 3 replicate sites in each of 2 crossed treatments for a total of 12 survey sites: (+MF) mass-flowering crops, (-MF) non-mass flowering crop, (+HR) adjacent hedgerow, (-HR) no adjacent hedgerow. Flower visitors were hand-netted for 1 hour of active search time, repeated 3-4 times in each bloom period (before, during, and after). We collected > 3000 bee specimens. *Osmia californica* nests were also deployed at sites for nutrition assays.).

Parasite loads (completed): Assess bee infection by parasites. using a random sample of approximately ten specimens per species per site (> 1600). Target pathogens included *Ascosphaera*, *Crithidia*, *Nosema* spp., and *Apicystis bombi*. We have completed the work, and published one paper so far, with an additional paper in review.

Cohen, H.*, Smith, G.P.*, Sardinias, H., Zorn, J., McFrederick, Q.M., Woodard, S.H., Ponisio, L.C. 2021. Mass-flowering crops attract bees, amplifying parasitism. *Proceedings B*, ([link](#))

Pathogen loads (completed): We assessed the pathogen (i.e., viral) load of our bees. We successfully screened 200 bees for 3 common RNA viruses (100% completed).

Bee microbiome (completed): We characterized the gut microbiota of adult bees in our study system through next-generation sequencing from the same bees screened for parasites. The extractions/sequencing are completed. We have two manuscripts about to be submitted.

Pollen meta-barcoding (completed). We removed a leg from the same bees screened for parasites/microbiome, and extracted the pollen DNA. We have completed the extractions/sequencing.

Pesticide exposure (completed): We are preparing a manuscript for submission.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

NA

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

1) Major activities: We accomplished fieldwork planning, fieldwork, specimen curation and identification. We spent over 300 hours doing field work in 2018-2019 with a hired crew of technicians and lab support. We collected over 4000 specimens total between the two years.

We have published two papers (Cohen et al. 2021a and Smith et al. 2021).

Cohen et al. 2021 summary: Monoculture farming is the cultivation of a single crop across fields measuring many thousands of acres. In California, the epicenter of monoculture farming, these systems are characterized by temporal gluts of floral resources in landscapes that are otherwise resource-poor. Many monoculture crops depend on the pollination provided by bees shipped in by beekeepers or naturally attracted to the mass-flowering event. Although it is widely agreed that the >20,000 species of bees found globally are declining, few studies address the impact of floral mass-blooms on disease dynamics in bees. We hypothesized that the aggregation of bees at mass-blooms enhances the spread of disease. We surveyed bee populations in monoculture sunflower farms and screened them for 7 parasites commonly implicated in bee declines. First, we discovered that some bee species were more vulnerable to parasitism than others (larger species who specialize on sunflower diets had higher parasitism rates), highlighting that one-size-fits-all conservation planning is likely ineffective with pollinators. Most importantly, we found that sites with historic legacies of sunflower cultivation are associated with more larger bee populations, and more bees are in turn linked with higher parasitism. There was a bright silver lining, that this relationship was actually reversed by growing non-crop, bee-friendly flowers alongside crop fields. We suggest that when growers install hedgerows, annual flowering strips, and enhance natural habitat in the landscape, bees disperse across landscapes, reducing transmission between infected individuals and leading to healthier populations.

Smith et al. 2021 summary: Sex-associated differences in behavior can have large ecological consequences, especially in plant-pollinator communities where floral visitor behavior affects plant reproduction. Whether these differences are prevalent enough to impact community-level processes, however, is unknown. Using 256 plant-pollinator communities, we built networks where the floral interactions of each sex were modeled separately, comparing observations to simulated networks where sex was randomized. We found that (1) in many species the sexes differed in their network roles and visited different partners, with females tending to visit more species and more peripheral species than males; (2) more generalist species differed more in network roles between the sexes; and (3) networks where nodes were separated by sex were more specialized than simulated networks, but were similarly resistant to perturbations. These findings suggest that despite variation among species, sex-associated differences in behavior are large enough to impact the network roles of male and female pollinators and common enough to influence the interaction patterns of entire plant-pollinator communities.

We have an additional paper that had been revised after review and is being re-submitted:

Cohen et al. submitted Proceedings B:

The microbiome and pathobiome are key determinants of how wildlife responds to global change, but it is unknown why microbial associations vary between and within species. We examine if host traits shape microbial associations at the species-level and individual-level. Using multilevel phylogenetic Bayesian models, we analyzed the microbiomes and parasites of 1,686 wild bees and managed honey bees across 34 species that vary widely in phenotypic

traits. We found that bees with more distinct traits host more distinct pathobiomes. For the microbiome, realized and fundamental diet breadth were a stronger predictor of microbial distinctness and diversity than trait distinctness. We also found that microbiome and pathobiome community composition were correlated, and that microbiome composition was further shaped by the community composition of the pollen collected by individual bees. Our findings demonstrate that both species-level and individual-level variation in host traits act as filters affecting the assembly of microbes in free-living organisms.

Two additional papers in manuscript:

Smith, G.P., Cohen, H., Zorn, J., McFrederick, Q.S., Ponisio, L.C.. Microbiome variability cannot be explained by plant-pollinator interaction networks alone. (expected submission Feb 2023, Ecology Letters)

Ponisio, L.C., Cohen, H., Zorn, J., Galbraith, S., Rivers, J. Beyond timber: harvested forests can support healthy, productive bee populations. (expected submission Jan 2023, Ecological Applications)

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

My lab group is committed to using the best practices in scientific computing and reproducible science. Our ultimate goal is to have all the materials needed to reproduce the study in its entirety (from data collection to analysis), available publicly and associated with all publications. We have:

- Curated all data in a relational database where automated data cleaning is conducted that is version controlled, including a record of any changes to the data and the raw data is never altered.
- Version controlled and post all protocols (collection, molecular, etc.) publicly, and associate each protocol DOI with its corresponding publication using protocols.io
- Version controlled all analytic code. Final products will be executable from a single script with an explanatory Rmarkdown file, publicly available and associated with corresponding publication (see Ponisio lab website for examples, <http://www.ponisiolab.com/reproducible-science.html>).
- Posted all final data publicly by the end of the grant period.

- 2.8. List key stakeholders that could be served by results of this project.

- Growers of pollinator-dependent mass-flowering crops (i.e., sunflower, almond, blueberry etc.)
- Beekeepers (Blue orchard bees, honey bees)

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Academic presentations:

FFAR Personnel	Position	Venue	Location	Title	Invited?	year	Format	type
Hamutahl Cohen	postdoc	Blue Orchard Bee Association meeting	Riverside, CA	Osmia californica for crop pollination in sunflower fields	yes	2020	talk	national
Hamutahl Cohen	postdoc	California Higher Education Sustainability conference	Santa Barbara, CA	Campus garden hedgerow to support healthy communities and farming ecosystems	No	2020	poster	state
Lauren Ponisio	PI	UCR grad recruitment	Riverside, CA	Pollinator resistance and interaction flexibility	yes	2020	talk	university
Lauren Ponisio	PI	UCR Policy and Agriculture	Riverside, CA	Promoting wild bee in CA Agriculture	yes	2020	talk	university
Lauren Ponisio	PI	ESA, ecological	virtual	Pollinator resistance and interaction flexibility	yes	2020	talk	national
Lauren Ponisio	PI	The Wildlife Society	virtual	Pollinator resistance and interaction flexibility	yes	2020	talk	national
Hamutahl Cohen	postdoc	Entomological Society of America	virtual	Mass-flowering crops attract bees, amplifying	yes	2020	talk	national
Hamutahl Cohen	postdoc	ESA, ecological	virtual	Mass-flowering crops attract bees, amplifying	no	2020	talk	national
Gordon Smith	postdoc	ESA, ecological	virtual	Sex-associated differences in the network roles of pollinators	no	2020	talk	national
Gordon Smith	postdoc	Blue Orchard Bee Association meeting	virtual	Osmia californica demography and disease in CA sunflower	yes	2020	talk	national
Lauren Ponisio	PI	META seminar	virtual	Mass-flowering crops attract bees, amplifying	yes	2020	talk	national



Lauren Ponisio	PI	UCLA EEOB dept seminar	virtual	Conservation and restoration of wild bee in CA Ag.	yes	2021	talk	department
Lauren Ponisio	PI	International Landscape Ecology Congress	virtual	Pollinator resistance and interaction flexibility	yes	2021	talk	international
Gordon Smith	postdoc	ESA, ecological	virtual	Microbiome variability cannot be explained by plant-pollinator interaction networks alone.	no	2021	talk	national
Lauren Ponisio	PI	Yale Biodiversity and Global Change Center Seminar	virtual	Host traits, microbes and disease in intensively managed agricultural landscapes	yes	2021	talk	department
Lauren Ponisio	PI	IEE seminar	virtual	Host traits, microbes and disease in intensively managed agricultural landscapes	yes	2021	talk	department
Lauren Ponisio	PI	Blue Orchard Bee Association meeting	virtual	Bee health in harvested forests	yes	2021	talk	national
Lauren Ponisio	PI	OIMB departmental seminar	Charelston, OR	Host traits, microbes and disease in intensively managed agricultural landscapes	yes	2022	talk	department
Lauren Ponisio	PI	Washington State	virtual	Host traits, microbes and disease in intensively managed agricultural landscapes	yes	2022	talk	department

Outreach to the public and land managers:

Led by	Position	Audience & intention	Event	Location, venue	Title	Date	Details
Hamutahl Cohen and lab	Postdoc	Increasing diversity in STEM	CSU Channel Islands Exchange	UCR		3/5/18	organized a visit from CSU channel islands including lab visits, Q&A with graduate students
Lauren Ponisio	PI	Increasing diversity in STEM	CAMP talk	UCR		4/4/18	https://highlanderlink.ucr.edu/organization/camp
Lauren Ponisio	PI	Increasing diversity in STEM	Persist talk	UCR	A tale of two bees	11/6/18	https://cnasscholars.ucr.edu/persist
Lauren Ponisio	PI	Increasing diversity in STEM	Panel: 7th Grade Girls in STEM	UCR		3/15/19	
Lauren Ponisio	PI	Public pollinator conservation outreach	PolliNation	podcast	Pollinator hedgerows	3/15/19	https://extension.oregonstate.edu/podcast/pollination-podcast/124-lauren-ponisio-pollinator-hedgerows



Lauren Ponisio	PI	Public pollinator conservation outreach	Pollinator Festival RCD	Land Use Learning Center	pollinator booth	11/2/19	https://www.rcrcd.org/2019-11-02-pollinator-festival-11am-3pm
Lauren Ponisio	PI	Increasing diversity in STEM	Persist talk	UCR	A tale of two bees	11/6/19	https://cnasscholars.ucr.edu/persist
Hamutahl Cohen and lab	Postdoc	Public pollinator conservation outreach	R Garden hedgerow installation	R garden		11/13/19	In Spring 2020, installed 150ft hedgerow at campus garden. Workshop attended by over 20 participants, including students and community members. Participants helped install hedgerow. Hands-on lecture component of event covered topics such as selection and maintenance of different hedgerow plants and promoting pollinators and other beneficial insects with diversified cropping systems. Presented this experience and tips for utilizing campus farms as a platform for outreach at the California Higher Education Sustainability Conference in Santa Barbara, CA in Summer 2020.
Lauren Ponisio	PI	Increasing diversity in STEM	Women in Science UCR panel	UCR		12/12/19	
Lauren Ponisio	PI	Public pollinator conservation outreach	Descanso Gardens Native bee day	Descanso gardens	booth	3/9/20	https://www.latimes.com/lifestyle/story/2020-01-16/art-embraces-tomatoes-apples-and-bees-at-descanso-gardens-fruiting-bodies-exhibit
Lauren Ponisio	PI	Public pollinator conservation outreach	CBS	television	Bee biologist cannot stop buzzing about her work	7/8/20	https://youtu.be/XsG6M-YdueQ
Lauren Ponisio	PI	Public pollinator conservation outreach	People Behind the Science	podcast	Working to Preserve and Restore Populations of Bees and Other Pollinators	7/20/20	http://www.peoplebehindthescience.com/dr-lauren-ponisio/
Lauren Ponisio	PI	Public pollinator conservation outreach	Wild World of bees	OSU extension virtual public seminar	The wild bees of agriculture	10/8/20	https://youtu.be/e06rVwaWsLA



Lauren Ponisio	PI	Public pollinator conservation outreach	NPR "All thing considered" interview	radio	Wildfires Open Forests for Wildlife and Research	1/26/21	https://www.klcc.org/post/wildfires-open-forests-wildlife-and-research
Lauren Ponisio	PI	Public pollinator conservation outreach	Environment for the Americas	webinar	Restoring habitat for pollinators	3/24/21	https://youtu.be/UmhKR4nvNDk
Lauren Ponisio	PI	Public pollinator conservation outreach	Jefferson Exchange	radio	Curious: bees in the trees (and elsewhere)	4/13/21	https://www.ijpr.org/show/the-jefferson-exchange/2021-04-09/mon-8-30-curious-bees-in-the-trees-and-elsewhere
Lauren Ponisio	PI	Land managers, pollinator conservation	California RCDS	webinar	Pollinators and Wildfires	5/28/21	https://youtu.be/V3BHV9fQ318
Lauren Ponisio	PI	Land managers, pollinator conservation	California Department of Fish and Wildlife	webinar	Wild bee conservation in California	9/9/21	https://us02web.zoom.us/j/7bzLjBkBPNUUIrUVEEKLuY1bAF2_35KetXlyzICNeSnDeZ_IGBWtntmg7RaA6vFzKuoqoVGG3biBO48.-x9noPKIBDf0ikRB?continueMode=true&x_zm_rtaid=nD9_f9nMTma9n_OK56-hrQ.1631641900889.97a8cb3ad865b8a5da9aec0fc46101cb&x_zm_rtaid=817
Lauren Ponisio	PI	Public pollinator conservation outreach	Jefferson Exchange	radio	Curious: improving working conditions for hardworking pollinators	10/11/21	https://www.ijpr.org/show/the-jefferson-exchange/2021-10-08/mon-8-30-curious-improving-working-conditions-for-hardworking-pollinators
Nicole Martinez	Grad student	Increasing diversity in STEM	Piney Point Elementary Gifted and Talented Friday Talks	webinar	The Chronicles of Pollination: The Plants, the Bugs and the food	11/5/21	40 min talk to underrepresented elementary (2nd to 5th grade) students in the gifted and talented program in Houston, Tx. Talk was about my journey and the importance of insects specially pollinators.
Lauren Ponisio	PI	Increasing diversity in STEM	UO Women in Stem	talk	A tale of two bees	4/13/22	
Lauren Ponisio	PI	Public pollinator conservation outreach	Eugene Natural History Society	webinar	Bees and fire!	4/15/22	
Lauren Ponisio	PI	Increasing diversity in STEM	SCORE	talk	A tale of two bees	5/17/22	

Lauren Ponisio	PI	Land managers, pollinator conservation	Pollinator Community of Practice	webinar	Bee, fire and clearcuts	20-Sep-22	
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- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

Our work was impacted strongly by COVID-19-related global supply chain related delays. With the approved extensions, we have been able to finish the proposed work. Because of the disruption some papers are still being submitted/ or are in revision.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

This project has provided employment for two postdoctoral scholars. At the start of employment, the postdoctoral scholars (with support from the primary PI) developed IDP documents to assess their skills in the following areas: scientific knowledge, research skills, communication, professionalism, management and leadership, research conduct, and career planning. The IDP plan was discussed with the PI to help identify the postdoctoral researcher's strengths and weaknesses. The postdocs/PI have also mentored undergraduate students on the research project, trained two laboratory technicians, and were given feedback on professional materials (such as CV, job application materials, and publications). The postdoctoral scholars have also participated in weekly project meetings with the Ponisio lab group and monthly project meetings with all PIs on the project and attended the weekly seminar series in the Biology department. *One postdoc was hired as an extension adviser at U Florida, the other was awarded an NSF postdoc fellowship.*

The two graduate students involved in the project from the Ponisio lab also completed IDPs with input from PI Ponisio and were trained in field and molecular techniques.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

4 field technicians

2 postdoctoral scholars
2 graduate student research assistantship
4 lab technicians
2 undergraduate assistantships

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

NA

- 3.2. Please provide a list of citations for the information products produced during the project.

Popular science:

Planting mixes of flowers around farm fields helps keep bees healthy, [The Conversation](#)

Academic:

Cohen, H.*, Smith, G.P.*, Zorn, J., McFrederick, Q.S., Ponisio, L.C., Host traits act as ecological filters shaping microbial community assembly (In review Proceedings B)

Cohen, H.*, Ponisio, L.C., Russell, K., McFrederick, Q.M., Philpott, S.M. Floral context determines parasite and pathogen outcomes for wild bees. *Molecular Ecology* ([link](#))

Smith, G.P.*, Gardner, J., Gibbs, J., Griswold, T., Hauser, M., Yanega D., Ponisio, L.C. 2021. Sex-associated differences in the network roles of pollinators. *Ecosphere* ([link](#))

Cohen, H.*, Smith, G.P.*, Sardinas, H., Zorn, J., McFrederick, Q.M., Woodard, S.H., Ponisio, L.C. 2021. Mass-flowering crops attract bees, amplifying parasitism. *Proceedings B*, ([link](#)) [pdf](#)

S. H. Woodard, S. Federman, R. R. James, B. N. Danforth, T. L. Griswold, D. Inouye, Q. S. McFrederick, L. Morandin, D. L. Paul, E. Sellers, J. P. Strange, M. Vaughan, N. M. Williams, M. G. Branstetter, C. Burns, J. Cane, A. B. Cariveau, D. P. Cariveau, A. Childers, C. Childers, D. L. Cox-Foster, E. C. Evans, K. K. Graham, K. Hackett, K. T. Huntzinger, R. E. Irwin, S. Jha, S. P. Lawson, C. Liang, M. M. López-Urbe, A. Melathopoulos, H. M.C. Moylett, C. Otto, L. C. Ponisio, L. L. Richardson, R. Rose, R. Singh, W. Wehling. Toward a U.S. national program for monitoring native bees. *Biological Conservation*. ([link](#))

Ponisio, L.C., Interaction flexibility and pyrodiversity increase pollinator population resistance, *Ecology and Evolution* ([link](#))

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

See above.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

<http://www.ponisiolab.com/restoring-pollinator-health.html>

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s)

NA

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

NA

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

We have created ID booklets for the bees found in sunflower fields, and sent those along with individualized reports on the bees found on each farm, and the book "Bees of California, Art and Poetry" (edited by postdoc Hamutahl Cohen) to each of the growers who collaborated with us.

In collaboration with UC Extension Specialist Wilson and RCD specialist Sardinas, we have ensured the results of this project and its implications for sunflower pollination and bee health are passed along relevant channels. We attended the Blue Orchard Bee Association meeting. Additionally, as part of his extension and outreach program CO-PI Wilson typically gives 10-20 talks per year at various grower and industry meetings focused on integrated pest management in orchard and vineyard crops (including almonds). Relevant findings from this

project will be presented at these meetings, with emphasis on the relationship between integrated pest management practices and pollinator health.

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

1. ~ 3000 individually identified, georeferenced bees/pollination insects from farms across Yolo Co.
2. >1000 parasite screenings of bees for 5 common parasite species
3. >1000 microbiome sequences of bees
4. pollen metabarcoding for > 1000 bees

5. Protein and lipid assays for *Osmia californica* nests

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

We have created separate databases for data collected from field conditions, including weather, specimens collection, site data, collection date, and data about who collected each specimen. We have also created a database of which analyses each specimen was used for (pollen identification, pesticide analysis, parasite testing, etc). Each dataset is a separate .csv file that was checked with R to ensure no typographic errors. A relational database of specimens is assembled using R protocols, which never transforming the original .csv files. This dataset is version controlled on Github to create a record of any changes made to the data.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

We have generated data in a relational database where automated data cleaning is conducted. All versions of analytic code are version controlled on Github (<https://github.com/lponisio/sunflower>). In addition as each study is published, the code relevant to that study is published in Zenodo with the manuscript.

Furthermore, we have developed protocols and posted them publicly on protocol.io which will be associated with a DOI.

Bee gut DNA extraction:

dx.doi.org/10.17504/protocols.io.q26g78k23lwz/v1

Bee pathogen extraction:

dx.doi.org/10.17504/protocols.io.36wgq4zwxvk5/v1

Bee pollen extraction:

dx.doi.org/10.17504/protocols.io.kxygxpbro18j/v1



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Certification

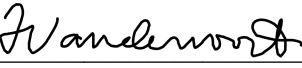
The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Lauren Ponisio

PI Signature:  Date: Nov 4, 2022

Authorized Signing Official (ASO) Name: Franny Vandervoort

ASO Signature:  Date: 11/09/2022